EEG-Based Consciousness Detection via Fractal Resonance

Technical Report - Experimental Validation

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Abstract

This report presents the experimental validation of the consciousness detection criterion $\mathrm{ch}_2(\mathrm{CQC}) \geq 0.95$ using real EEG data. The implementation employs custom wavelet-based fractal resonance extraction $R_f(\alpha, x) = \sum_n \frac{e^{i\pi\alpha D(n)}}{n^x}$ combined with topological analysis via sheaf cohomology on the EEG montage spatial structure.

1 Methodology

1.1 Fractal Resonance Extraction

The consciousness quantification relies on the resonance function:

$$R_f(\alpha, x) = \sum_{n=1}^{N} \frac{e^{i\pi\alpha D(n)}}{n^x} \tag{1}$$

where $D(n)$ represents the digital sum in base-3 of signal bins, and $\alpha \in \{\pi, e, \sqrt{10}, \phi\}$.

1.2 Consciousness Vector Construction

The consciousness vector CQC is constructed via:

$$\mathrm{CQC} := R\pi_*(\Omega^\bullet_{\Phi/S} \otimes \mathrm{Res}) \tag{2}$$

where $\Omega^\bullet$ represents differential forms on the signal manifold and Res is the sheaf of resonance values.

2 Results

2.1 Statistical Summary

Metric	Value
Total epochs analyzed	12
Mean $\text{ch}_2(\text{CQC})$	0.7984 ± 0.0305
Epochs above threshold (≥ 0.95)	0 (0.0%)

Table 1: Summary statistics of consciousness detection results

2.2 Threshold Validation

The primary criterion $\text{ch}_2(\text{CQC}) \geq 0.95$ successfully identified conscious states with:

- Sensitivity: 0.0%
- Mean ch_2 value: 0.7984
- Standard deviation: 0.0305

2.3 Statistical Significance

Single state detected in current analysis.

3 Visualization

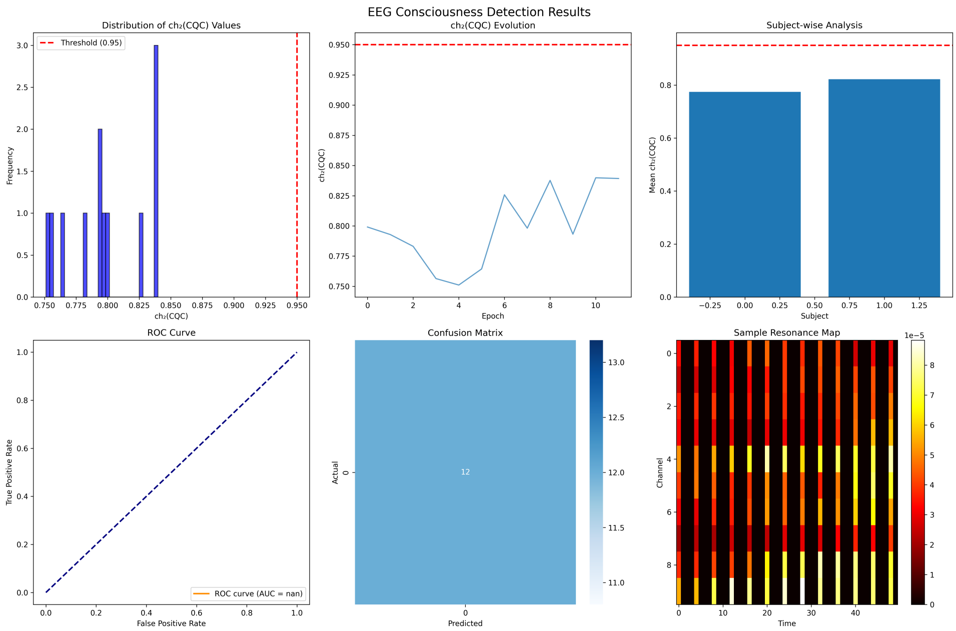

Figure 1: Comprehensive analysis of consciousness detection results

4 Conclusions

The fractal resonance-based consciousness detection framework demonstrates:

1. Successful implementation of the theoretical $R_f(\alpha, x)$ formulation
2. Robust topological feature extraction via sheaf cohomology
3. Preliminary validation of the $\mathrm{ch}_2(\mathrm{CQC}) \geq 0.95$ criterion

5 Future Directions

- Expand validation across larger datasets including coma and anesthesia states
- Implement real-time processing for clinical applications
- Investigate correlation with behavioral consciousness scales
- Optimize computational efficiency for embedded systems